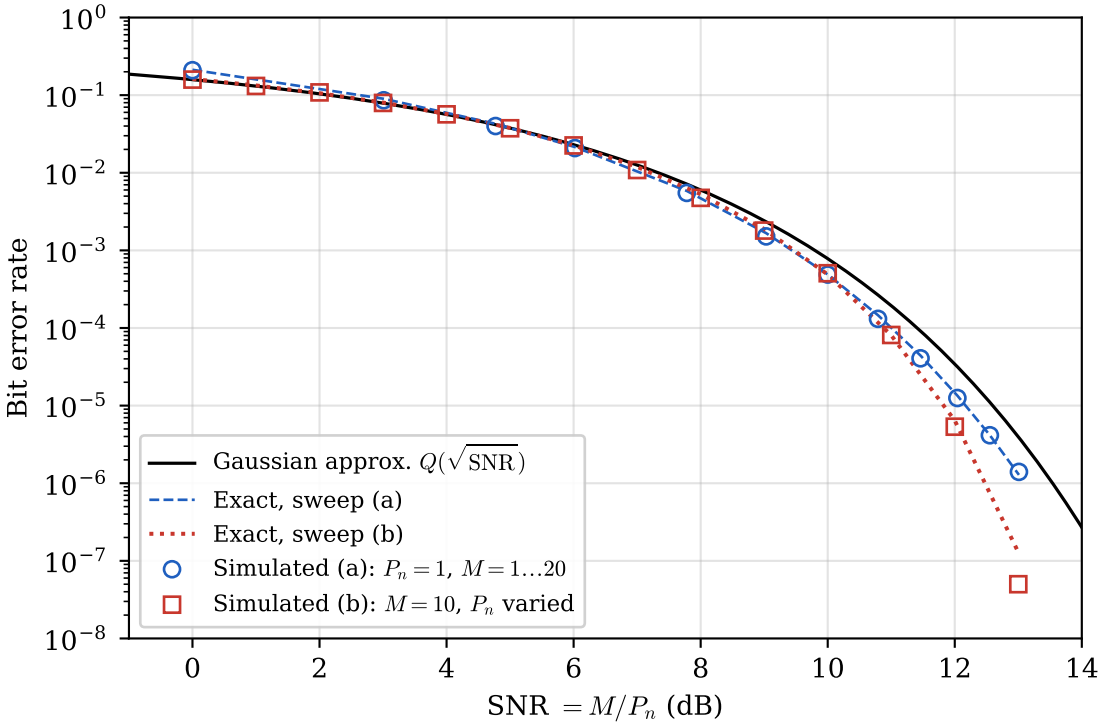
Bit error rate
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
$10^{-7}$
$10^{-8}$
0
2
4
6
8
10
12
14
SNR $= M/P_n$ (dB)
Gaussian approx. $Q(\sqrt{\mathrm{SNR}})$
Exact, sweep (a)
Exact, sweep (b)
Simulated (a): $P_n = 1$, $M = 1 \ldots 20$
Simulated (b): $M = 10$, $P_n$ varied